

Project Asteria - Detailed Notes (Sessions 8 & 9)

Learning Engine

The Learning Engine converts experiences into improved behaviour. Unlike the Memory Manager, which simply stores information, the Learning Engine extracts patterns from repeated experiences. Every experience contains the situation, action, result, outcome and confidence. Asteria stores every observation initially with very low confidence instead of deciding immediately whether it is important. Repeated confirmation increases confidence, contradictory evidence decreases it, and lack of observations over time causes gradual confidence decay. Rather than deleting old knowledge, memories transition through Active, Dormant and Archived states so rare but important experiences can be recalled later. Source reliability also matters: multiple agreeing sensors or trusted human input increase confidence faster than uncertain observations.

Planning, Mission Management and Prediction

The Planning Engine breaks large goals into smaller tasks and evaluates alternative plans using energy, risk, time and mission reward. Plans can be repaired dynamically when the environment changes. Above it sits the Mission Manager, which maintains active missions, pauses or resumes them and decides which mission should be planned next. During discussion we improved this design by introducing feasibility alongside priority. A high-priority mission should not always be attempted immediately if the probability of success is extremely low. Instead, each mission is evaluated using priority, urgency, required resources, expected success probability, consequences of failure and alternative plans. To support this, a Prediction Engine mentally simulates future outcomes before execution, allowing Asteria to choose actions with the highest expected chance of overall mission success.